AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST224
Mathematical Methods

Final Examination
Spring Semester 2025/2026

Date: 3 /June/2026

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Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 3H

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

PART 1

1. [5×2 marks] Choose the correct answer:

1.1 Let $\mathbf{A} = x^2 z\,\mathbf{i} - 2y^3 z^2\,\mathbf{j} + xy^2 z\,\mathbf{k}$, then $\nabla \cdot \mathbf{A} =$

a) $2xz\,\mathbf{i} - 6y^2 z^2\,\mathbf{j} + xy^2\,\mathbf{k}$.
b) 0.
c) $2xz^3\mathbf{j} + 6xy^2 z\mathbf{k}$.
d) $2xz - 6y^2 z^2 + xy^2$.
e) None of the above.

1.2 The general solution of the differential equation $y' = 5y$, is:

a) $y = C_1 \cos 5x - C_2 \sin 5x$
b) $y = C\, e^{5x}$
c) $y = C_1 \cos 5x + C_2 \sin 5x$
d) $y = C_1 e^{5x} + C_2 e^{-5x}$
e) None of the above.

1.3 Let $f(x, y) = \frac{x}{y-6}$, then $\frac{\partial f}{\partial x}$ at $(2,1)$ is:

a) $-\frac{2}{5}$
b) 0
c) $-\frac{1}{5}$
d) $\frac{1}{5}$
e) None of the above.

1.4 The length of the quarter-circle path by using the parametric equations $x(t) = 2\cos t$, $y(t) = 2\sin t, (0 \le t \le \pi/2)$, is

a) π.
b) $\pi/2$.
c) 2π.
d) $\pi/4$
e) None of the above.

1.5 If $\nabla\phi = 2xyz^3\,\mathbf{i} + x^2 z^3\,\mathbf{j} + 3x^2 yz^2\,\mathbf{k}$, then the possible form of $\phi(x, y, z)$ is:

a) $x^2 yz^3 + 20$.
b) $x^2 yz^3\,\mathbf{i} + x^2 yz^3\,\mathbf{j} + x^2 yz^3\,\mathbf{k}$.
c) $yz^3\,\mathbf{i} + x^2 z^3\,\mathbf{j} + 3x^2 y\,\mathbf{k}$.
d) $x^2 y^2 z^3$.
e) None of the above.

2. [5×2 marks] Determine whether following statements are True or False and justify your answer in both cases.

2.1 $\int_1^2 \int_3^4 x^2 e^y\, dy\, dx = \int_1^2 x^2 dx \int_3^4 e^y\, dy$

2.2 If $\mathbf{F}$ is a vector field, then $\nabla \times \mathbf{F}$ is a vector field.

2.3 If $f(x, t) = \cos(3x - 2t)$, then $f_{xx} = f_{tt}$.

2.4 The function $f(x, y) = x^2 y - x^2 - \frac{1}{3}y^3$ has a stationary point at $(1,1)$.

2.5 The differential equation $y' = xy^2$ is a linear first-order differential equation

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PART 2

Q1: [3 marks] the position vector of a point P is given by

$$\mathbf{r}(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k} \quad for\ 0 \le t \le 2.$$

i. Find the velocity $\mathbf{v}(t)$ and the acceleration $\mathbf{a}(t)$ of P at time t.

ii. Find $\mathbf{v}(1)$ and $\mathbf{a}(1)$.

Q2: [4 marks] Consider the function

$$z = x^2 - 3y^2 + 20, \quad x = 2\cos t\ , \quad y = 2\sin t.$$

Find $\frac{dz}{dt}$ at $t = \frac{\pi}{4}$.

Q3: [4 marks] Solve the initial value problem

$$\frac{dy}{dx} = \frac{x+y}{x}\ , \qquad x > 0, \quad y(1) = 2.$$

Q4: [4 marks] Let $f(x,y) = x^3 + y^3$

a) Calculate the slope of the function $f(x,y)$ at the point $(1,1)$, in the direction of the (non-unit) vector $\mathbf{i} - \mathbf{j}$.

b) Prove that $\nabla \times \nabla f = \mathbf{0}$.

PART 3

Q1: [5 marks] Assume that the solution of a partial differential equation is

$$u(x,t) = A\sin(x)\,e^{-t},$$

where A is a constant.

i. Show that $u(x,t)$ satisfies the heat equation $u_t = u_{xx}$.

ii. Determine the value of A if the initial condition is:

$$u(x,0) = 5\sin x.$$

Q2: [4 marks] Use the divergence theorem to calculate the surface integral of the vector field

$$\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$$

over the surface V of a unit cube with vertices $(0,0,0)$ and $(1,1,1)$.

Q3: [6 marks] Consider the non-linear system of differential equations

$$\dot{x} = x(1-y)\ ,$$
$$\dot{y} = y(x-2).$$

i. Find the equilibrium points of the system.

ii. Find the Jacobian matrix of the system.

iii. In the neighborhood of each equilibrium point:

- find the linear system of differential equations that gives the approximate behavior of the system near the equilibrium points.
- find the eigenvalues of the Jacobian matrix.
- use the eigenvalues to classify the equilibrium point of the linearized system.

End of questions

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